NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2015-2016

MH1812 - DISCRETE MATHEMATICS

May 2016 TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **ZERO (0)** questions and comprises **ZERO (0)** printed pages.

2. Answer **ALL** questions. The marks for each question are indicated at the end of each question.

3. Answer each question beginning on a **FRESH** page of the answer book.

4. This **IS NOT** an **OPEN BOOK** exam.

5. Candidates may use calculators. However, they should write down systematically the steps in the workings.

QUESTION 1.

(a) Compute $5^{2016} \mod 7$. **(8 marks)**

(b) Determine whether $(\neg q \wedge (p \to q)) \to \neg p$ is a tautology. (list all intermediate propositions if you choose to prove by truth table) **(10 marks)**

(c) How many solutions are there for the following equation

$$x_1 + x_2 + \cdots + x_r = n$$

with r, n, x_i positive integers for $i = 1, 2, \ldots, r$ and $n \geq r$. **(10 marks)**

(d) Let sets $A = \{1, 3\}$, $B = \{0, 2, 4\}$, and $P(x, y)$ denote "$5 \mid (x + y)$", i.e., $(x + y)$ is multiple of 5. Determine the truth value of the following and justify your answer: **(12 marks)**

- $\forall x \in A, \exists y \in B, P(x, y)$.
- $\exists y \in B, \forall x \in A, P(x, y)$.

QUESTION 2.

(a) Find the solution of the recurrence relation, $a_n = 4a_{n-1} - 4a_{n-2}$, with $a_1 = 2$ and $a_2 = 8$. **(10 marks)**

(b) Prove

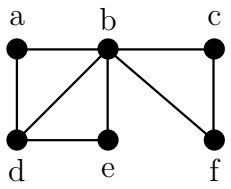
$$1 \cdot 2 + 2 \cdot 3 + \cdots + n(n + 1) = n(n + 1)(n + 2)/3$$

for every positive integer n. **(10 marks)**

QUESTION 3.

(a) Let A, B, and C be sets, show $(B - A) \cup (C - A) = (B \cup C) - A$. **(10 marks)**

(b) Refer to the graph below, find Euler Path, Euler Circuit and Hamilton Circuit if any, justify your answer if it does not exist. **(8 marks)**



QUESTION 4.

(a) Let set $A = \{a, b, c, d\}$ and relation $R = \{(a, a), (a, b), (b, c), (c, d), (d, c)\}$. **(12 marks)**

- Is R reflexive, symmetric, transitive ?
- Find R^t, i.e., the transitive closure of R.

(b) Let function $f(x) = x^2 + 2x + 3$ with x being real numbers and $x \leq -1$, find **(10 marks)**

- the range of f.
- the inverse function f^{-1}.

END OF PAPER